

2.1 Vector-Valued Functions

1. Find the domain of the following vector functions.

a) $\mathbf{r}(t) = \left\langle \sqrt{16-t^2}, \ln(t+3), \frac{1}{1-t} \right\rangle.$

b) $\mathbf{r}(t) = \frac{t-2}{t+2} \mathbf{i} + \sin(t) \mathbf{j} + \ln(9-t^2) \mathbf{k}$

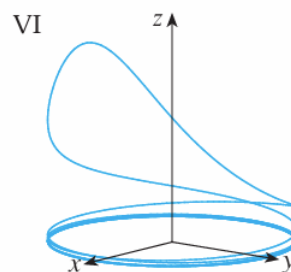
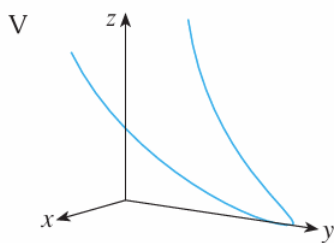
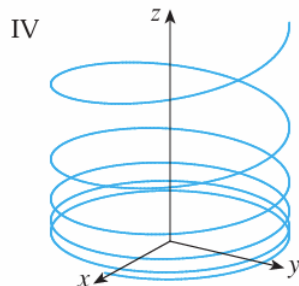
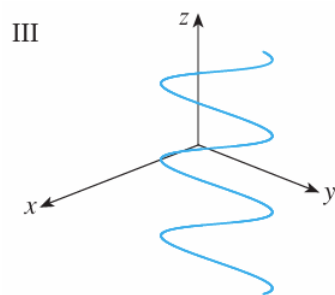
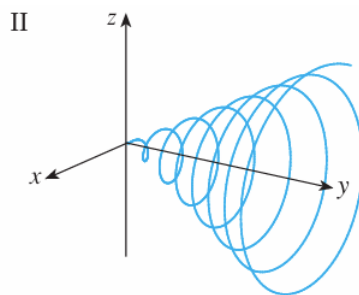
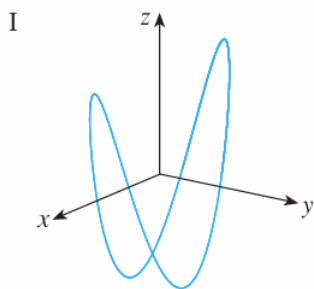
c) $\mathbf{r}(t) = \left\langle \frac{1}{t-2}, \sqrt{t+3}, \cos(t) \right\rangle$

d) $\mathbf{r}(t) = \left\langle \ln(t^2-4), \frac{1}{\sqrt{t}}, t^3 \right\rangle$

e) $\mathbf{r}(t) = \arcsin(2t-1)\mathbf{i} + \sqrt{t}\mathbf{j} + e^{-t}\mathbf{k}$

2. Find any three different parameterizations of the graph $y = x^2$ on the interval $1 \leq x \leq 4$.
3. Find any three different parameterization of the graph of $y = \ln(x^2) + 2x$ on the interval $2 \leq x \leq 5$
4. Find any three different parameterization for a single tracing of the graph of $x^2 + y^2 = 4$.

5. Match the parametric equations with the graphs below. Give reasons for your choices.



a) $\mathbf{r}(t) = \langle t \cos(t), t, t \sin(t) \rangle$ for $t \geq 0$.

b) $\mathbf{r}(t) = \left\langle \cos(t), \sin(t), \frac{1}{1+t^2} \right\rangle$

c) $\mathbf{r}(t) = \left\langle t, \frac{1}{1+t^2}, t^2 \right\rangle$

d) $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + \cos(2t)\mathbf{k}$

e) $\mathbf{r}(t) = \cos(8t)\mathbf{i} + \sin(8t)\mathbf{j} + e^{0.8t}\mathbf{k}$ for $t \geq 0$.

f) $\mathbf{r}(t) = \cos^2(t)\mathbf{i} + \sin^2(t)\mathbf{j} + t\mathbf{k}$

6. True or False: The functions $\mathbf{r}(t) = \langle \cos(3t), \sin(3t) \rangle$ and $\mathbf{s}(t) = \langle 3\cos(t), 3\sin(t) \rangle$ generate the same curve.

7. Sketch the curve with the given vector equation. Indicate with an arrow the direction in which t increases.

a) $\mathbf{r}(t) = \langle \sin(t), t \rangle$

b) $\mathbf{r}(t) = \langle t^3, t^2 \rangle$

c) $\mathbf{r}(t) = \langle t, 2-t, 2t \rangle$

d) $\mathbf{r}(t) = \langle \sin(\pi t), t, \cos(\pi t) \rangle$

e) $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j} + 2\mathbf{k}$

f) $\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(t)\mathbf{j} + 2\mathbf{k}$